

that can store a mere billion billion bytes in such a volume; and massively parallel computers of the same size that can deliver a billion billion instructions per second.

- ▶ More powerful and smaller computers will encrypt our data and provide round-the-clock security.

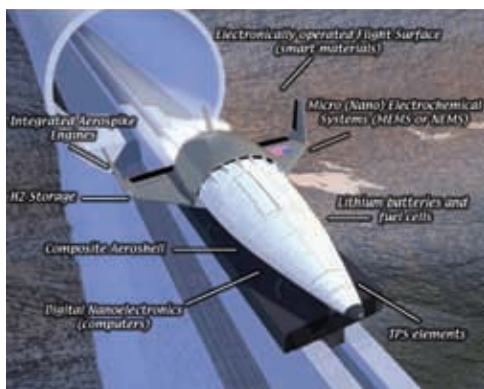
Quantum cryptography – cryptography that utilises the unique properties of quantum mechanics – will provide unbreakable security for businesses, government and military. These same quantum mechanics will be used to construct quantum computers capable of breaking current encryption techniques.



Small is the way forward

Transportation

- ▶ Today, most airplanes are made from metal despite the fact that diamond has a strength-to-weight ratio over 50 times that of aerospace aluminum. Nanotechnology will let us inexpensively make shatterproof diamond



Future transportation to zip you there faster

(with a structure that might resemble diamond fibres) in the exact shapes we want. This would let us make a Boeing 777 whose unloaded weight is 50 times lighter but just as strong.

- ▶ Today, travel in space is very expensive and reserved for an elite few. Nanotechnology will dramatically reduce

the costs and increase the capabilities of spaceships and space flight. It will also provide extremely powerful computers with which to guide both, those ships and a wide range of other activities in space. Mars, Jupiter, and someday, Andromeda, all within reach!